

PROFESSIONAL SUMMARY

Full Stack Developer and AI/ML Engineer with expertise in building high-performance web applications, scalable backend systems, and automated AI-driven pipelines. Proficient in Node.js, React, PHP, Python, Java, and MySQL. Experienced in designing and deploying ML/DL models, Computer Vision solutions, and data analytics workflows that improve efficiency, enhance user experience, and enable intelligent decision-making. Strong problem solver with a track record of delivering measurable impact.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, PHP, C/C++, SQL, HTML, CSS, Bash
- **Web Development:** Node.js, Express.js, React.js, Flask, Spring Boot, RESTful APIs, Bootstrap, EJS, AJAX
- **Machine Learning / AI:** Machine Learning, Deep Learning, Computer Vision, CNN, RNN, TensorFlow, Keras, Scikit-learn, Transfer Learning (MobileNet, ResNet, VGG16), Model Training, Hyperparameter Tuning, Optimization, Data Preprocessing, Feature Engineering, Evaluation and Classification Metrics
- **Data Analytics / Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- **Tools / Version Control:** Git, GitHub, VS Code
- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, OOP, DBMS, Software Engineering
- **Soft Skills:** Problem Solving, Self-Learning, Presentation, Adaptability

PROFESSIONAL EXPERIENCE

- **Software Developer — Automation and Data Analytics** Apr 2025 – Sep 2025
Zuxtra Network Pvt. Ltd. Kolkata, India
 - Engineered automation systems using Python, integrating ML-driven content recommendations and AI-assisted post scheduling, increasing operational efficiency by 40%.
 - Developed analytics modules to detect engagement patterns and predict high-performing posting windows, boosting campaign effectiveness by 35%.
 - Built scalable RESTful APIs with PHP and Python for AI inference workflows, reducing data latency by 25%.
 - Designed optimized MySQL schemas and ETL pipelines for ML data ingestion, resulting in 30% faster query and model-processing performance.
 - Applied NLP techniques for caption and hashtag analytics, generating high-relevance content recommendations.
- **Software Engineer — Blood Donation Management System** Jul 2022 – Jul 2023
Euphoria GenX Kolkata, India
 - Developed modules for donor registration, request tracking, and data pipelines, improving operational efficiency by 35%.
 - Integrated analytics to monitor donor history, blood availability, and demand trends for insights-driven decision-making.
 - Built Spring Boot REST APIs for scalable data exchange in ML workflows, improving backend speed by 25%.
 - Implemented robust data validation workflows optimized for predictive modeling requirements.
 - Designed optimized MySQL schemas for analytical queries, achieving a 20% reduction in response time.

PROJECTS

- Brain Hemorrhage Detection using Deep Learning

Jan 2024 – May 2025

M.Tech Project: Automated CT scan analysis using supervised and deep learning models

- Achieved 96% accuracy using CNN and transfer-learning models (MobileNet, ResNet, VGG16).
 - Performed preprocessing, augmentation, feature engineering, and resolved overfitting/ generalization issues.
 - Built a Flask-based web application for real-time CT scan prediction, reducing diagnostic latency by 40%.
 - Evaluated ML models: Logistic Regression, KNN, SVM, Naive Bayes, Decision Trees, Random Forest, XGBoost, LightGBM, CatBoost.
- Habit Tracker and NLP Analytics

Nov 2023 – Feb 2024

Full-stack application integrating NLP and ML for habit prediction and insights

- Applied NLP to analyze user text and generate personalized habit recommendations.
 - Built supervised ML models for habit prediction, trend detection, and behavior classification.
 - Implemented secure authentication using Passport.js and Express sessions; designed scalable REST APIs.
 - Utilized MongoDB with Mongoose for structured data modeling and efficient querying.
 - Tech Stack:** Python, TensorFlow, Keras, Node.js, Express.js, MongoDB, React.js, JavaScript, HTML, CSS, Bootstrap.

EDUCATION

- M.Tech, Computer Science and Engineering

2025

Kalyani Government Engineering College, Kalyani

CGPA: 6.96
- B.Tech, Computer Science and Engineering

2023

Dr. Sudhir Chandra Sur Institute of Technology, Kolkata

CGPA: 8.85
- Senior Secondary (Science)

2018

Gitaram Academy, Berhampore, West Bengal

Grade: 6.9
- Matriculation

2016

Kendriya Vidyalaya, Berhampore, West Bengal

Grade: 6.6

CERTIFICATIONS

- Full Stack Web Development

Coding Ninjas

2023
- Career Camp: Data Structures in Java

Coding Ninjas

2023
- Martian Internship Program 2022

Persistent Systems Limited

2022

HOBBIES

- Gaming

FPS and strategy games (BGMI, Free Fire, PC gaming); enhances reflexes, decision-making, and problem-solving.
- Cooking

Experimenting with new recipes and cuisines; builds creativity, patience, and attention to detail.

LANGUAGES

English	Full Professional
Hindi	Native / Bilingual
Bengali	Native / Bilingual